



**EXPLAINABLE MULTIMODAL MODEL FOR CHEST X-RAY DISEASE
CLASSIFICATION USING IMAGES AND RADIOLOGY REPORTS**

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ABSTRACT

Deep learning has demonstrated strong performance in chest X-ray disease classification using image data alone, yet most approaches ignore the complementary diagnostic information in radiology reports. Multimodal learning addresses this by integrating both modalities, but existing systems rarely provide transparency into how predictions are formed across modalities, limiting clinical trust and usability. This study developed and evaluated an explainable multimodal framework for multi-label chest X-ray disease classification using the MIMIC-CXR dataset, while investigating the impact of late and joint fusion strategies and generating interpretable predictions across both modalities.

DenseNet121 and LLaMA 3.1 8B were used as image and text encoders respectively, and combined using late and joint fusion. Grad-CAM and Integrated Gradients were applied for explainability, with joint explanations derived from the jointly trained fusion model. Multimodal fusion models achieved stronger performance than single-modality models. Late fusion achieved the highest macro AUROC of 0.9238 and sensitivity, while joint fusion achieved higher F1-score, precision, and specificity. Statistical testing confirmed significant performance differences between all models ($p < 0.05$), although per-class analysis showed limited significant differences between fusion strategies. Explainability analysis demonstrated clinically meaningful attributions from both modalities, with joint explanations providing consistent cross-modal evidence.

Overall, combining chest X-ray images and radiology report text improves disease classification, while the two fusion strategies demonstrated different strengths across evaluation metrics. Clinically interpretable explanations were generated for both image and text predictions, to enhance transparency in AI decision-making and support trust in clinical settings.

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1. INTRODUCTION AND BACKGROUND

1.1 INTRODUCTION

Chest radiography remains a fundamental diagnostic tool in clinical practice, widely used to detect conditions such as pneumonia, atelectasis, pleural effusion, and cardiomegaly. Despite its importance, interpretation is challenging, with error rates of around 4% in routine practice and up to 30% in abnormal cases, translating to millions of potential errors annually (Gefter et al., 2023). Growing imaging demand and a global shortage of radiologists make reliable, interpretable automated systems increasingly urgent (Bhayana, 2024).

While artificial intelligence has significantly improved automated chest X-ray analysis, most existing approaches focus only on image data, often ignoring the accompanying radiology reports which contain valuable clinical information and diagnostic reasoning (Johnson et al., 2019). Multimodal fusion addresses this by integrating both modalities within a unified framework, and prior studies have shown it can improve predictive performance through complementary information (Upadhyaya et al., 2024).

Multimodal fusion more closely reflects how clinicians combine image findings with radiology reports during diagnosis (Huang et al., 2020). However, most existing studies use only one fusion strategy, leaving the impact of different fusion mechanisms on multimodal learning and prediction behaviour insufficiently understood. Beyond performance, interpretability is a critical requirement in medical AI because models must provide transparent, clinically meaningful explanations to support clinician trust and decision-making (Abbas et al., 2025).

This study addresses these limitations by developing an explainable multimodal framework for chest X-ray disease classification, combining chest X-ray images and radiology report text within a single architecture. The framework uses DenseNet121 as the image encoder and LLaMA 3.1 8B as the text encoder, combined using late fusion and joint fusion strategies. Explainability is incorporated through Grad-CAM for image predictions and Integrated Gradients for text predictions, with joint analysis across both modalities.

1.2 AIMS AND RESEARCH QUESTIONS

This study aims to develop an explainable multimodal framework for chest X-ray disease classification using the MIMIC-CXR dataset, by integrating chest X-ray images and radiology report text. Specifically, the framework investigates the impact of late and joint fusion strategies on multimodal learning, while incorporating explainability across both modalities to support transparent and clinically meaningful predictions.

To achieve this aim, the following research questions are addressed:

RQ1: Does combining image and text modalities improve chest X-ray disease classification performance over single-modality models?

RQ2: How do late and joint fusion strategies influence multimodal disease classification performance?

RQ3: Can clinically interpretable explanations be generated across both image and text modalities to support transparent predictions?

2. RELATED WORK

2.1 UNIMODAL CHEST X-RAY DISEASE CLASSIFICATION

Deep learning has greatly improved chest X-ray disease classification, particularly through convolutional neural networks (CNNs) for automated image analysis. CNN-based models have demonstrated strong performance in identifying thoracic abnormalities directly from chest X-ray images by learning hierarchical visual features without manual feature engineering. One of the most influential studies in this area was CheXNet by Rajpurkar et al. (2017), which used a DenseNet121 architecture for pneumonia detection and achieved performance comparable to radiologists on the ChestX-ray14 dataset. The success of CheXNet demonstrated the potential of deep learning for assisting chest X-ray interpretation and inspired numerous subsequent studies.

More recently, Khater et al. (2025) developed AttCDCNet, an attention-enhanced convolutional neural network for chest disease classification from chest X-ray images, aiming to improve feature extraction and disease localisation through attention mechanisms. Their work demonstrated the effectiveness of attention-guided image modelling for thoracic disease detection. Similarly, Pillai (2022) applied DenseNet-based classification to the MIMIC-CXR dataset and reported an AUROC of 0.78 across multiple thoracic diseases, further demonstrating the suitability of DenseNet architectures for chest X-ray analysis. However, these approaches relied solely on image data despite radiology reports containing valuable clinical observations and diagnostic reasoning that may provide additional information beyond what is visible in the image alone (Danu et al., 2023).

2.2 MULTIMODAL CHEST X-RAY DISEASE CLASSIFICATION

The integration of chest X-ray images with accompanying radiology reports represents a natural extension of unimodal approaches, motivated by the clinical reality that radiologists interpret both visual evidence and textual clinical context simultaneously.

Recent studies have explored multimodal learning by combining chest X-ray images with textual clinical information. Multimodal learning can be implemented through several fusion strategies, each combining

information from different modalities at different stages of the learning process. Early fusion combines raw input features or low-level representations before any modality-specific processing, allowing the model to learn joint representations from the earliest processing stage. Late fusion integrates predictions from separate unimodal models at the decision level, enabling each modality to be processed independently before combining outputs. Joint fusion, also referred to as intermediate or feature-level fusion, combines intermediate representations from each modality before the final classification decision. This approach enables the model to learn cross-modal feature interactions that may not be accessible when modalities are combined only at the decision level. More recently, transformer-based fusion approaches have gained attention by employing attention mechanisms and cross-modal interactions to model complex relationships between visual and textual information. The choice of fusion strategy can significantly influence model performance, interpretability and the extent to which complementary information from different modalities is utilised.

Upadhyaya et al. (2024) investigated multimodal fusion for chest X-ray disease classification using chest X-ray images and radiology reports from the MIMIC-CXR dataset. Their framework employed ResNet50 and BERT as image and text encoders respectively and compared both early and late fusion architectures for multi-label classification. The study reported that the late fusion model achieved an accuracy of 94.2%, outperforming the early fusion model (91.4%) and the image-only model (81.9%), while the text-only model achieved the highest accuracy of 96.1%. In addition, the ROC curves presented in the study suggest approximate AUC values of 0.927, 0.756, 0.617 and 0.492 for the text-only, late fusion, early fusion and image-only models respectively, indicating superior discrimination performance for the text-only and late fusion approaches. However, the study relied primarily on accuracy as its main evaluation metric despite the highly imbalanced nature of chest X-ray datasets, where accuracy may provide an overly optimistic assessment of model performance. Furthermore, no explainability analysis was provided, limiting insight into how visual and textual information contributed to model predictions.

Jacenków et al. (2022) proposed a multimodal transformer framework that combined chest X-ray images with indication text extracted from radiology reports. Unlike most multimodal approaches that utilise full radiology reports, the study incorporated only patient information available prior to imaging, such as clinical history and examination indications. Image features extracted using ResNet-50 were integrated with textual representations generated by BERT and evaluated on the MIMIC-CXR dataset. Their proposed Multimodal Bitransformer (MMBT) achieved a micro-AUROC of 87.8%, outperforming both unimodal baselines, including BERT (81.7%) and ResNet-50 (84.4%), as well as a previous multimodal approach (86.0%). The results demonstrated that incorporating clinically relevant textual information can improve chest X-ray disease classification performance, although, the framework evaluated only a single transformer-based fusion strategy.

Similar work by Jahanian et al. (2025) proposed RadFuse, a transformer-based multimodal framework that jointly analysed chest X-rays and radiology reports using Vision Transformer (ViT) and ClinicalBERT encoders connected through cross-attention layers for joint diagnosis and report retrieval. The framework was evaluated on the MIMIC-CXR and IU X-ray datasets, achieving an AUC of 0.847 on MIMIC-CXR for the disease classification task. However, explainability was limited to attention visualisations, which may not always provide reliable explanations for model decisions.

Liu et al. (2025) proposed MDFormer, a transformer-based multimodal framework for chest disease classification that combined chest X-ray images, radiology reports and physiological measurements within a unified architecture. The model employed modality-specific encoders and a dedicated fusion mechanism to capture interactions across modalities, enabling more comprehensive clinical representation than unimodal approaches. Their work achieved a macro AUROC of 0.9028 and outperformed several baselines, demonstrating the benefits of integrating complementary information from multiple clinical modalities for disease classification. However, the study focused on single fusion architecture, and the contribution of individual modalities was not systematically investigated.

2.3 EXPLAINABILITY IN MEDICAL AI

The increasing adoption of artificial intelligence in healthcare has raised important concerns regarding transparency, accountability and clinical trust. Unlike many commercial applications, medical AI systems often support decisions that may directly influence patient diagnosis and treatment. As a result, clinicians are frequently reluctant to rely on predictions generated by models that cannot provide understandable justifications for their decisions. Abbas et al. (2025) emphasised that explainability is essential for the safe and effective integration of AI into clinical workflows, as healthcare professionals require evidence that model predictions are based on clinically meaningful features rather than spurious correlations or dataset biases.

To address these concerns, a wide range of explainable artificial intelligence (XAI) techniques have been developed. One of the most influential contributions in this area was Gradient-weighted Class Activation Mapping (Grad-CAM) proposed by Selvaraju et al. (2017), which generates visual explanations by highlighting image regions that contribute most strongly to a model's prediction. Grad-CAM has since become one of the most widely adopted explainability techniques in medical imaging because it enables clinicians to assess whether a model is focusing on anatomically relevant structures and pathological regions. Other approaches, including Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP), provide feature-level explanations by estimating the contribution of individual inputs to model predictions (Ribeiro et al., 2016; Lundberg and Lee, 2017). These methods have been successfully applied across a range of healthcare applications to improve transparency and facilitate model auditing.

Despite these advances, most explainability research in medical AI has focused on single-modality systems, particularly image-based disease classification models. Existing studies commonly employ Grad-CAM to identify clinically relevant image regions, yet such approaches provide little insight into how textual information contributes to predictions when multiple modalities are used. This limitation becomes increasingly important as multimodal frameworks continue to gain popularity within medical AI. When predictions are generated from both images and clinical text, understanding the contribution of each modality is essential for assessing model reliability and supporting clinical decision-making. Motivated by these challenges, this study incorporates multimodal explainability through the combined use of Grad-CAM for visual interpretation and Integrated Gradients for textual interpretation. This enables investigation of both image regions and report terms that contribute to disease predictions, providing greater transparency into how multimodal information influences model decision-making.

2.4 RESEARCH GAP

Despite the promising results achieved by multimodal approaches, several limitations remain. Existing studies have largely evaluated a single fusion strategy, making it difficult to determine how different fusion mechanisms influence multimodal learning and disease classification performance. As a result, there is limited understanding of the relative strengths and weaknesses of alternative approaches such as late fusion, joint fusion and transformer-based fusion. Furthermore, explainability remains underexplored, with many studies prioritising predictive performance while providing limited insight into how visual and textual information contribute to model decisions. Consequently, there remains a need for interpretable multimodal frameworks that not only compare alternative fusion strategies within a consistent experimental setting, but also provide meaningful explanations across both image and text modalities. This study addresses these gaps by evaluating multiple fusion approaches, including late fusion and joint fusion, while incorporating Grad-CAM and Integrated Gradients to improve transparency and interpretability in chest X-ray disease classification.

3. METHODOLOGY

3.1 DATASET DESCRIPTION

This study utilised the MIMIC-CXR-JPG dataset (Johnson et al., 2019), a large publicly available chest radiography dataset comprising 377,110 chest X-ray images across 227,827 studies from 64,588 patients collected at Beth Israel Deaconess Medical Center. Each study includes a chest X-ray image, a corresponding radiology report, and disease labels generated using the CheXpert labelling framework (Irvin et al., 2019). Images, radiology reports and CheXpert labels were merged using subject and study identifiers to ensure accurate pairing of each study with its corresponding image and report.

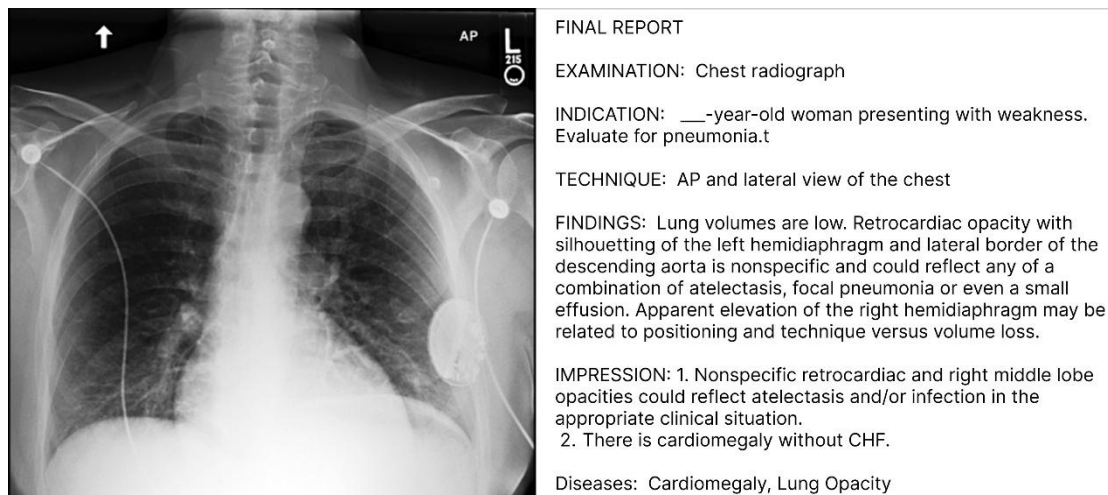


Figure 1: MIMIC-CXR dataset sample showing a chest X-ray image alongside the corresponding report sections with associated CheXpert disease labels.

The radiology reports contain several sections, the indication section states the clinical reason for the examination and relevant patient history, while Findings describe the radiologist's observations of the image, and the Impression section contains a summary of the radiologist's final diagnostic assessment. The dataset also contains multiple image view positions, including posteroanterior (PA), anteroposterior (AP), lateral and oblique views.

Due to storage and computational constraints, a subset of 59,246 studies was selected for analysis. Despite this reduction, the dataset remained sufficiently large for deep learning model development and is comparable to previous studies using partial MIMIC-CXR subsets (Holste et al., 2024). Like many real-world clinical datasets, MIMIC-CXR exhibits missing values and substantial class imbalance, where certain thoracic diseases occur far less frequently than others, presenting challenges for model training (Holste et al., 2024).

3.2 DATA PREPROCESSING

3.2.1 DATA CLEANING AND PREPARATION

A study-level approach was adopted since CheXpert labels are defined at the study level, making image-level duplication inappropriate (Johnson et al., 2019). Only frontal chest X-rays were retained for analysis. PA views were selected where available due to their higher diagnostic quality and reduced magnification effects; otherwise, AP views were used.

The impression section of each radiology report was excluded because it contains direct diagnostic summaries that closely correspond to the target disease labels and could therefore introduce leakage and artificially inflate performance. Consequently, only the Indication and Findings sections were retained as textual inputs. Reports were cleaned using rule-based methods to remove formatting artefacts while preserving clinical terminology and report structure (Danu et al., 2023). Studies missing both Indication and Findings sections were excluded.

Labels exceeding 80% missing values were removed as they provided insufficient positive samples for stable learning, and Support Devices was excluded because it does not represent a disease, leaving eight target labels (Holste et al., 2024). Uncertain and missing labels were mapped to negative using the U-Zeros strategy, which maximises training data utilisation but may introduce label noise for ambiguous cases (Irvin et al., 2019). No learned preprocessing was applied before dataset splitting to prevent information leakage.

The dataset was split at the subject level into training (70%), validation (10%) and test (20%) sets comprising 42,008, 5,745 and 11,493 studies respectively. A subject-level split ensured that studies from the same patient could not appear across multiple partitions, thereby preventing patient-level leakage and providing a more realistic evaluation of model generalisation. Class weights were computed from the training set only to avoid bias.

3.2.2 IMAGE AND TEXT PREPROCESSING

Chest X-ray images were converted from grayscale values (0–255) to three-channel Red, Green and Blue (RGB) format and resized to 224×224 pixels to meet the input requirements of ImageNet-pretrained convolutional neural networks (Irvin et al., 2019). During training, images were resized to 256 pixels, randomly cropped to 224×224 , converted to tensors, and normalised using ImageNet mean and standard deviation statistics. Random cropping was applied to improve model robustness by exposing the network to slight spatial variations during training while preserving diagnostically relevant image content. Horizontal flipping was intentionally excluded because reversing anatomical structures such as the heart can produce clinically unrealistic chest radiographs and potentially introduce systematic prediction errors. During validation and testing, deterministic preprocessing was applied using image resizing, centre cropping to 224×224 , tensor conversion and identical normalisation to ensure consistent model evaluation.

The Indication and Findings sections were extracted from each report. Following extraction, the report text was cleaned and the selected sections were combined into a single input sequence. The text was tokenised using the pretrained LLaMA 3.1 8B tokenizer implemented through the Hugging Face AutoTokenizer. Tokenisation converts raw text into subword units that can be processed by the language model while preserving clinical terminology and domain-specific vocabulary. Reports were truncated or padded to a

maximum sequence length of 512 tokens to produce uniform inputs for batch processing. Although LLaMA 3.1 8B supports context windows of up to 128,000 tokens, a limit of 512 tokens was selected for computational efficiency because the clinically relevant information within radiology reports is typically contained within concise structured sections such (Danu et al., 2023).

Unlike traditional natural language processing approaches that construct a vocabulary directly from the training corpus, LLaMA employs a pretrained tokenizer developed from large-scale text corpora. Consequently, no dataset-specific vocabulary was learned during preprocessing, reducing the risk of information leakage and enabling the model to leverage rich semantic representations learned during pretraining. This approach allows clinically related terms and observations to be represented in a contextual manner, where the meaning of a word depends on its surrounding text rather than being treated as an isolated token.

3.3 EXPLORATORY DATA ANALYSIS (EDA)

EDA was conducted on the pre-processed dataset of 59,246 studies to examine label distributions and inter-label relationships prior to modelling.

Label Distribution

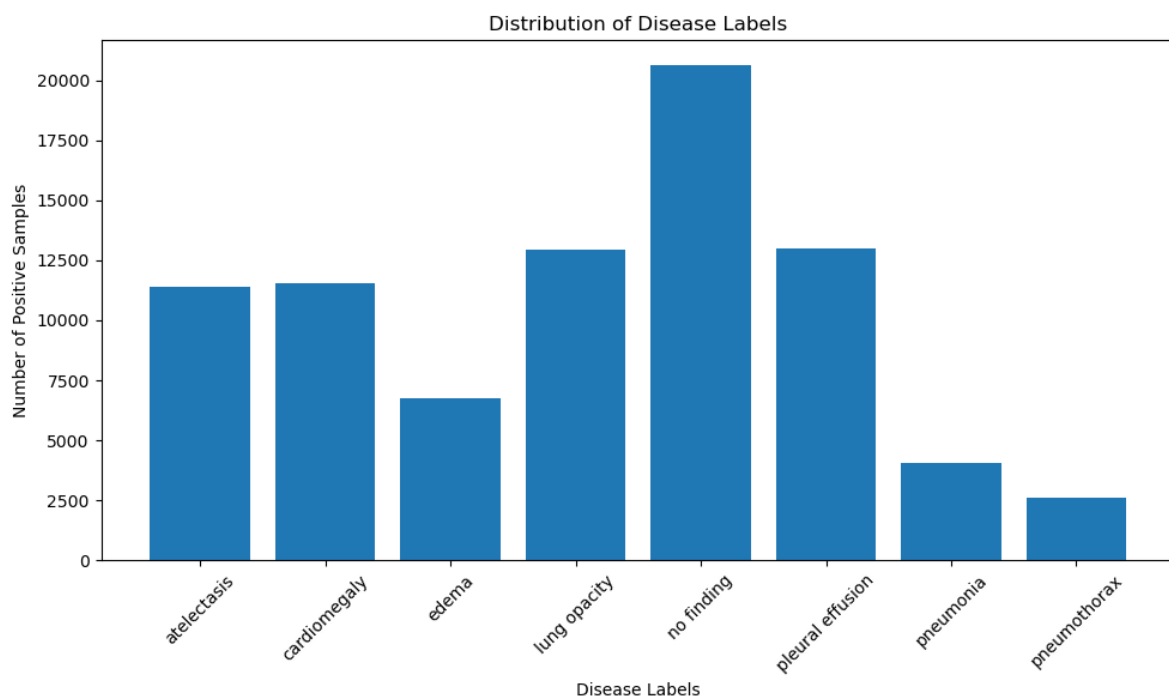


Figure 2: Distribution of positive samples across the selected 8 disease labels. No finding represents no disease.

Pneumonia and pneumothorax are highly underrepresented.

The distribution of positive samples revealed substantial class imbalance. No finding, pleural effusion, and lung opacity were the most frequent, while pneumonia and pneumothorax were severely underrepresented, consistent with the long-tailed distribution characteristic of MIMIC-CXR (Holste et al., 2024). This motivated the use of class-weighted binary cross-entropy loss to penalise misclassification of minority classes.

Disease Co-occurrence

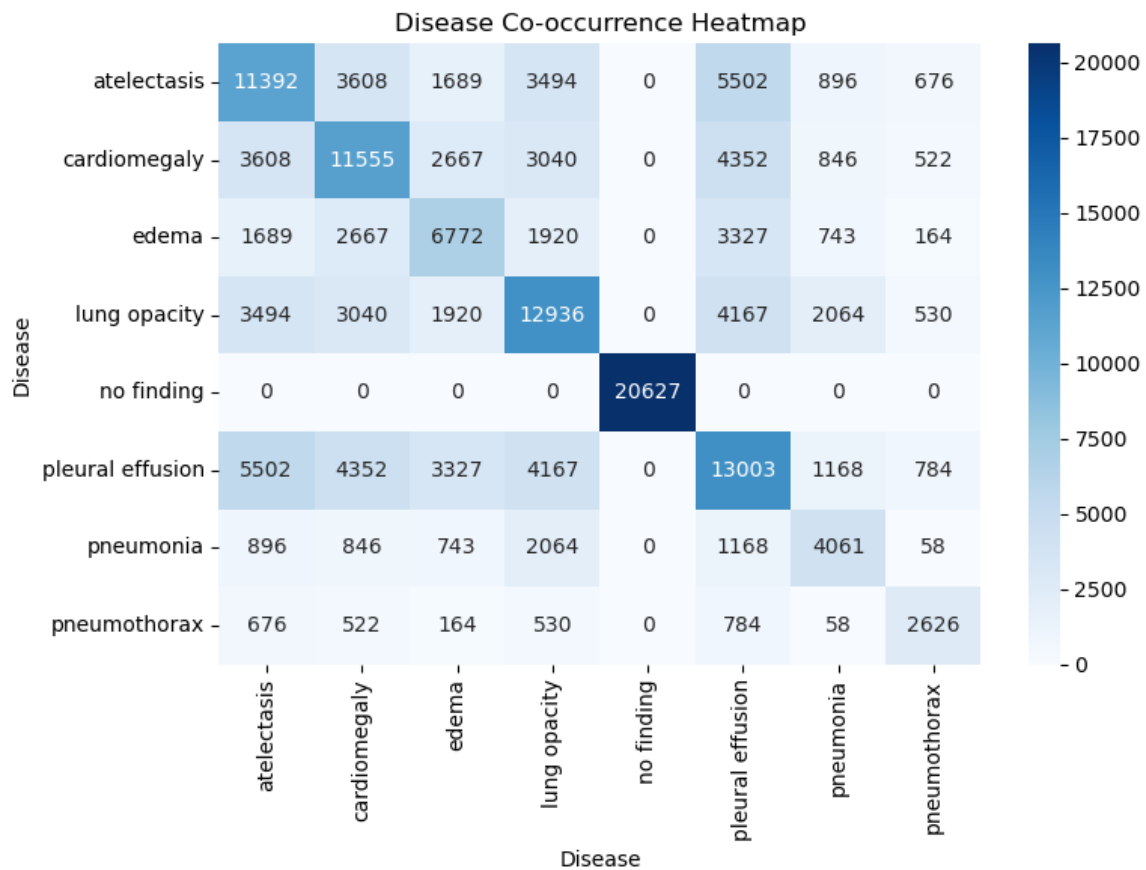


Figure 3: Disease co-occurrence matrix showing the frequency with which pairs of diseases appear together in the same imaging study.

Co-occurrence analysis revealed clinically meaningful relationships, with pleural effusion frequently co-occurring with atelectasis and cardiomegaly, and lung opacity appearing alongside atelectasis and pleural effusion. The no finding label showed no co-occurrence with any disease label, as expected. These patterns confirm the multi-label nature of the task (Holste et al., 2024).

3.4 MODEL ARCHITECTURE

DenseNet121 was selected as the image encoder because its densely connected architecture promotes feature reuse and efficient gradient propagation, enabling effective learning of subtle and spatially distributed radiological findings (Huang et al., 2017). Compared with deeper architectures such as ResNet, DenseNet achieves competitive performance with fewer parameters while reducing feature redundancy through dense connectivity. Furthermore, DenseNet121 has been widely adopted in chest X-ray analysis, including the CheXNet framework (Rajpurkar et al., 2017), demonstrating its suitability for thoracic disease classification. ImageNet-pretrained weights were used to provide robust feature initialisation, and the architecture is well suited to Grad-CAM visualisation due to its convolutional feature maps. The final classification layer was replaced with a linear head producing eight class outputs, with no sigmoid activation applied since BCEWithLogitsLoss incorporates this internally.

LLaMA 3.1 8B was selected as the text encoder due to its strong contextual language understanding and large-scale pretraining on diverse text corpora, enabling rich semantic representations of clinical narratives. Unlike encoder-only models such as BERT and BioClinicalBERT, LLaMA leverages the representational capacity of modern large language models to capture complex relationships between clinical observations and disease concepts. To enable computationally efficient fine-tuning, four-bit quantisation was applied using Quantized Low-Rank Adaptation (QLoRA) (Dettmers et al., 2023), while Low-Rank Adaptation (LoRA) modules were inserted into the query and value projection layers, reducing trainable parameters to 3.44 million (0.046% of the full model). Token embeddings were aggregated using mask-aware mean pooling across the sequence, allowing information to be captured from the entire report rather than relying on a single token representation (Reimers and Gurevych, 2019). A linear classification head was added for eight-class prediction. However, mean pooling may dilute highly informative clinical terms by averaging them with less relevant tokens, potentially reducing sensitivity to disease-specific observations.

3.5 MODEL TRAINING

DenseNet121 was trained for 20 epochs with a batch size of 16, dropout rate of 0.3, AdamW optimiser, learning rate of 1×10^{-4} and weight decay of 1×10^{-4} . BCEWithLogitsLoss with class-specific positive weights was used to address class imbalance (Holste et al., 2024), while mixed-precision training improved computational efficiency. Early stopping with a patience of 5 epochs based on validation macro AUROC was applied to reduce overfitting, and the best-performing checkpoint was retained for evaluation.

LLaMA 3.1 8B was trained for 3 epochs using AdamW with a learning rate of 2×10^{-4} , batch size of 1 and gradient accumulation to maintain stable optimisation under GPU memory constraints. Three epochs were

considered sufficient given the strong pretrained representations already encoded within the model. All models were trained using GPU acceleration.

Hyperparameter tuning was conducted using a one-at-a-time (OAT) strategy, where each parameter was varied independently while others were held constant (Jiang, 2021). This approach was selected because exhaustive search methods such as grid search and Bayesian optimisation would have been computationally expensive, particularly for LLaMA 3.1 8B fine-tuning. Learning rate, dropout, batch size and number of epochs were explored for DenseNet121, while tuning for LLaMA was limited to learning rate, batch size and dropout due to its computational demands. Model selection was performed using validation macro AUROC, and the best-performing configuration was used for final training. Although computationally efficient, OAT does not account for interactions between hyperparameters and may therefore overlook globally optimal parameter combinations.

Training was conducted using Python, PyTorch and the Hugging Face Transformers library on an NVIDIA A100 GPU (40 GB VRAM) with CUDA acceleration.

3.6 MULTIMODAL FUSION

Multimodal learning can be implemented through different fusion strategies depending on the stage at which information from each modality is combined. Late fusion integrates modality-specific predictions after independent processing, whereas joint fusion combines learned feature representations before classification, enabling direct interaction between modalities. Both approaches were investigated in this study to evaluate how the level of multimodal integration influences disease classification performance and model interpretability.

3.6.1 LATE FUSION

Late fusion combines the predicted probabilities of independently trained image and text models using a weighted average, as illustrated in Figure 4:

$$P_{fused} = \alpha \cdot P_{image} + (1 - \alpha) \cdot P_{text}$$

where P_{image} and P_{text} are the predicted probabilities from DenseNet121 and LLaMA 3.1 8B respectively, and $\alpha \in [0,1]$ is a fusion weight optimised on the validation set by maximising macro F1-score. This strategy treats each modality as an independent expert and combines their predictions only at the decision level. As a result, each encoder can specialise in learning modality-specific information without being influenced by the other

modality. Late fusion can be computationally efficient because it does not require retraining of unimodal models. In addition, modality-specific predictions are preserved, making the contribution of each modality easier to interpret. Furthermore, the independent nature of the architecture makes it robust to situations where one modality contains noisy or incomplete information. However, because fusion occurs only after prediction, the model cannot learn cross-modal interactions between image and text features, potentially limiting its ability to exploit complementary information distributed across modalities (Huang et al., 2020).

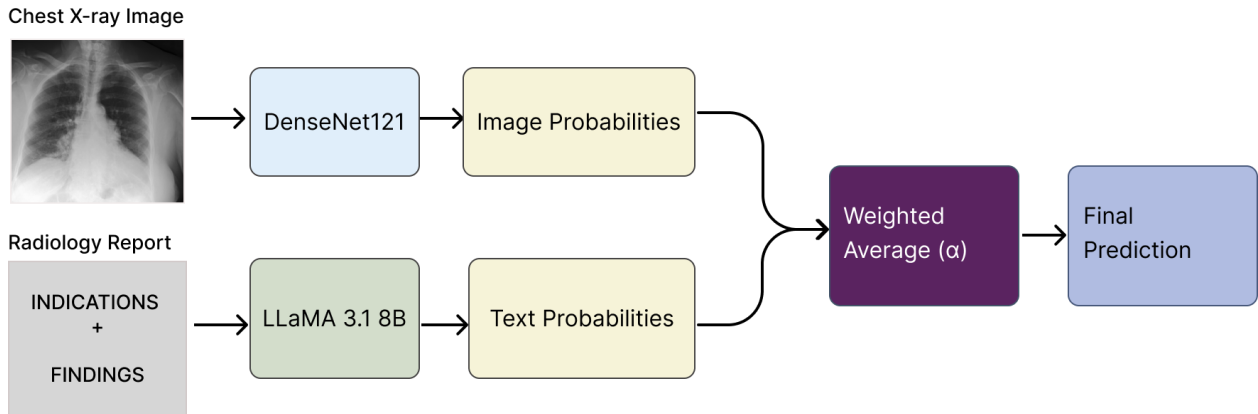


Figure 4: Simple late fusion architecture combining image and text predictions.

3.6.2 JOINT FUSION

Joint fusion enables deeper multimodal integration by learning from image and text representations simultaneously. DenseNet121 and LLaMA 3.1 8B were used as parallel encoders to generate image and text feature embeddings respectively. These embeddings were concatenated into a shared multimodal representation and passed through a fully connected classification head that produced eight disease predictions, as illustrated in Figure 5.

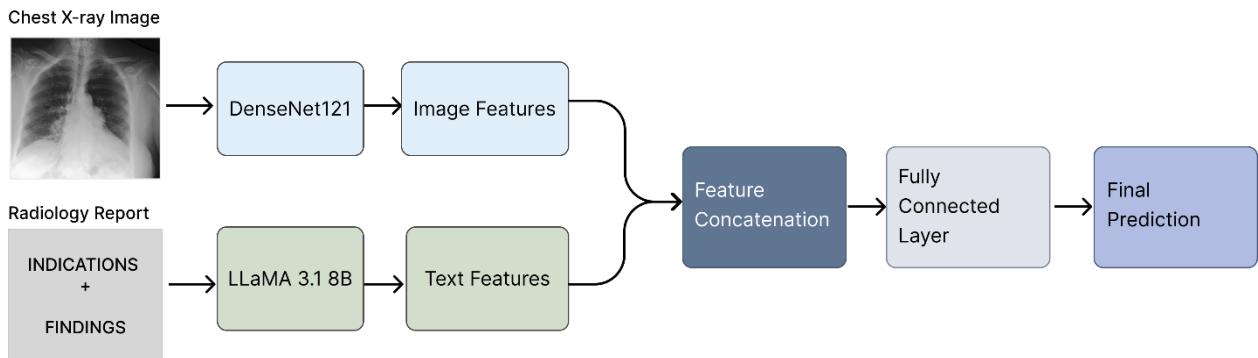


Figure 5: Simple joint fusion architecture integrating image and text features.

Unlike late fusion, which combines final predictions, joint fusion integrates information at the feature level, allowing relationships between visual and textual representations to be learned during training. This enables the model to capture complementary evidence across modalities, such as radiological observations described in reports that correspond to abnormalities identified within chest X-ray images. The shared representation is optimised through end-to-end backpropagation, allowing gradients from the classification objective to influence both encoders simultaneously and encourage multimodal feature alignment. Consequently, joint fusion has the potential to learn richer representations than decision-level approaches. However, the stronger interaction between modalities increases model complexity and may reduce interpretability, as the individual contribution of each modality becomes less explicit after feature integration.

3.7 EVALUATION METRICS

Class-specific thresholds were optimised on the validation set by selecting values that maximised F1-score for each label and were fixed before test evaluation to prevent contamination. Model performance was primarily assessed using macro-averaged AUROC, which is threshold-independent and robust to class imbalance (Irvin et al., 2019). Macro averaging was adopted to ensure that each disease contributed equally to the overall performance estimate regardless of class frequency. F1-score, sensitivity, specificity and precision were additionally reported as macro averages. Accuracy was excluded because it can be misleading under severe class imbalance (Holste et al., 2024), and the test set was used strictly once. Statistical significance between model pairs was assessed using bootstrap testing for macro AUROC comparison and DeLong’s test for per-class AUROC analysis on the same test set.

3.8 EXPLAINABILITY

Gradient-based explainability methods were applied after evaluation due to their direct relationship with model predictions and established attribution properties (Sundararajan et al., 2017). Two stages were implemented: unimodal explainability and multimodal explainability.

For unimodal explainability, Grad-CAM was applied to DenseNet121 to generate spatial heatmaps highlighting the image regions that most influenced each prediction. Grad-CAM was selected because it produces class-specific visual explanations directly from convolutional feature maps without requiring architectural modifications, making it particularly suitable for medical image interpretation (Selvaraju et al., 2017). Integrated Gradients was applied to LLaMA 3.1 8B to compute token-level importance scores by estimating how strongly each token influenced the predicted output relative to a baseline input (Sundararajan et al., 2017). Integrated Gradients was selected because it provides theoretically grounded

feature attributions while avoiding the extensive sampling and perturbation procedures commonly used by model-agnostic approaches such as LIME and SHAP, making it more computationally efficient for large language models.

For multimodal explainability, both methods were applied within the joint fusion model, with Grad-CAM applied to the image encoder and Integrated Gradients to the text encoder. Explanations were derived from a single forward pass through the jointly trained architecture, ensuring that attributions reflected the fused model rather than independent unimodal behaviour (Imran, Lee and Lee, 2025).

4. RESULTS

This section presents the model performance results, followed by statistical and explainability analysis. Hyperparameter tuning improved performance over baseline training for DenseNet121 model, with optimized configurations achieving higher validation metrics. The best configuration was selected for final evaluation on the test set. The performance of the models was evaluated using macro-averaged metrics (see Appendix B for additional hyperparameter tuning details).

4.1 MODEL PERFORMANCE

Table 1: Performance comparison of unimodal and multimodal models using macro-averaged metrics (Bold values indicate the highest scores)

Model	Modality	AUROC	F1	Precision	Sensitivity	Specificity
DenseNet121	Image	0.8172	0.4932	0.4147	0.6251	0.8194
LLaMA 3.1 8B	Text	0.9177	0.6460	0.6152	0.7018	0.8966
Late Fusion	Image + Text	0.9238	0.6565	0.6035	0.7272	0.8986
Joint Fusion	Image + Text	0.9201	0.6769	0.6384	0.7251	0.9122

Table 1 presents the macro-averaged performance of all four models on the test set. DenseNet121 achieved a macro AUROC of 0.8172, establishing the image-only baseline. LLaMA 3.1 8B substantially outperformed DenseNet121 across all evaluation metrics, achieving a macro AUROC of 0.9177. This improvement of over 10 percentage points suggests that the radiology reports contained highly informative diagnostic information for disease classification.

Both multimodal models further improved performance beyond the unimodal baselines. Late Fusion achieved the highest macro AUROC (0.9238) and sensitivity (0.7272), while Joint Fusion recorded the highest F1-score (0.6769), precision (0.6384) and specificity (0.9122). Although the performance differences between the two fusion strategies were small, both consistently outperformed the image-only model and achieved higher AUROC than the text-only model, indicating that combining image and report information provided complementary predictive value. This trend is reflected in the ROC curves shown in Figure 5, where the multimodal models cluster together and closely follow LLaMA while remaining clearly above DenseNet121 across all false positive rates. However, the improvements over the text-only model were relatively modest compared with the substantial performance gap observed between the image-only and text-only models.

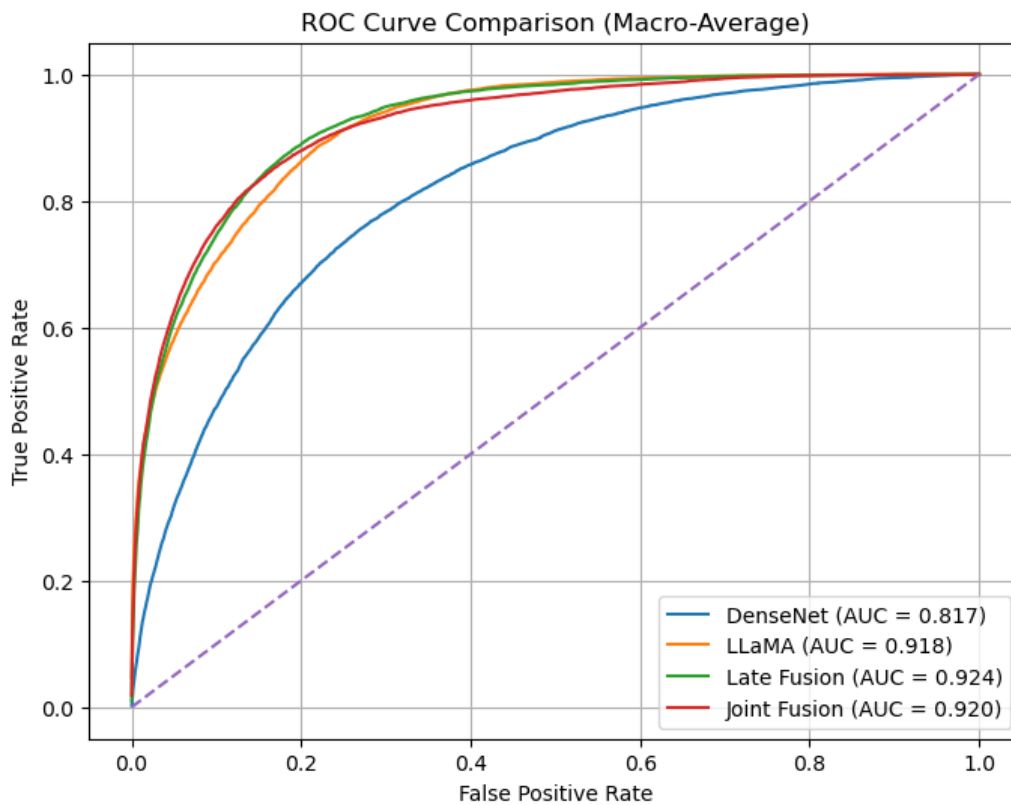


Figure 6: ROC curve showing the unimodal and multimodal models.

Table 2: Per-class AUROC comparison of multimodal models (Late and Joint fusion) and unimodal models (LLaMA 3.1 and DenseNet121) across eight disease labels

Disease	DenseNet121 (AUROC)	LLaMA 3.1 8B (AUROC)	Late Fusion (AUROC)	Joint Fusion (AUROC)
Atelectasis	0.8077	0.8928	0.9048	0.9008
Cardiomegaly	0.7919	0.9052	0.9145	0.9111
Edema	0.8838	0.9391	0.9501	0.9510
Lung opacity	0.7584	0.9043	0.9066	0.9049
No finding	0.8553	0.9322	0.9345	0.9333
Pleural effusion	0.8955	0.9349	0.9478	0.9512
Pneumonia	0.7177	0.8626	0.8667	0.8465
pneumothorax	0.8269	0.9706	0.9657	0.9624
Macro Average	0.8172	0.9177	0.9238	0.9201

Per-class AUROC results in Table 2 show that DenseNet121 performed well on pleural effusion (0.8955) and edema (0.8838) but struggled with pneumonia (0.7177) and lung opacity (0.7584), where class imbalance reduces positive training samples. LLaMA 3.1 8B outperformed DenseNet121 across all disease classes, achieving the highest per-class AUROC for pneumothorax (0.9706) and outperforming the fusion models on this label.

Both fusion models consistently outperformed the unimodal baselines across disease labels. Late fusion achieved the highest AUROC in five classes: atelectasis, cardiomegaly, lung opacity, no finding, and pneumonia, while joint fusion led in edema, pleural effusion. This shows that late fusion consistently outperformed at the per-class level.

4.2 ABLATION STUDY

The results in Table 1 were analysed to examine how each modality and fusion strategy contributed to overall performance. DenseNet121 alone achieved a macro AUROC of 0.8172. Using text alone (LLaMA 3.1) improved macro AUROC to 0.9177. Introducing multimodal fusion further improved performance over both baselines. Late fusion gained 0.0061 AUROC over LLaMA, while joint fusion gained 0.0024, confirming that

combining both modalities provides additional predictive value beyond text alone. Notably, joint fusion achieved higher F1, precision and specificity compared to late fusion, suggesting that feature-level integration produces more precise and balanced predictions, while late fusion achieved higher AUROC and sensitivity. These results confirm that each component; image, text, and fusion strategy, contributes meaningfully to overall performance, and that radiology reports provided the dominant predictive signal, while image features contributed complementary information.

4.3 STATISTICAL SIGNIFICANCE ANALYSIS

Table 3: Bootstrap-Based Statistical Comparison of Model Macro AUROC Performance with 95% Confidence Intervals and p-values.

Comparison	AUROC Diff	CI Lower	CI Upper	p-value	Significant
LLaMA 3.1 8B vs DenseNet121	0.101	0.096	0.105	0.002	Yes
Late Fusion vs LLaMA 3.1 8B	0.006	0.005	0.007	0.002	Yes
Joint Fusion vs LLaMA 3.1 8B	0.002	0.0004	0.005	0.022	Yes
Late Fusion vs Joint Fusion	0.004	0.002	0.006	0.001	Yes

Statistical testing confirmed that the differences in macro AUROC across models were statistically significant ($p < 0.05$), suggesting that the performance differences observed were not random, as seen in Table 3. The largest difference was observed between the text and image models (AUROC diff=0.101, $p=0.002$). Both fusion models showed significant improvements over the text model, while late fusion significantly outperformed joint fusion (diff=0.004, $p=0.001$).

Per-class analysis using DeLong's test revealed limited statistically significant differences between the fusion strategies despite differences in overall macro AUROC. Although Late Fusion achieved higher AUROC for most disease classes, statistically significant differences were observed in only two classes: Pleural Effusion and Pneumonia ($p < 0.05$). Joint Fusion achieved higher AUROC for Pleural Effusion, while Late Fusion performed better for Pneumonia. The remaining disease classes showed no statistically significant differences, indicating broadly comparable performance between the two fusion approaches at the individual disease level. Detailed pairwise statistical comparisons are provided in Appendix A.

4.4 EXPLAINABILITY ANALYSIS

Explainability analysis was applied to DenseNet121 and LLaMA3.1 8B for image and text predictions, with joint multimodal explanations generated for selected test cases. Figure 6 shows the heatmap for a cardiomegaly case, where activation is concentrated around the cardiac region and central thoracic area. Activation is primarily located within clinically relevant structures rather than irrelevant background regions. Token importance scores were computed for text explainability using Integrated Gradients for a sample radiology report. Figure 7 shows the top 15 tokens ranked by attribution score. Highly ranked tokens included clinically meaningful terms such as cardiomeastinal, consolidation, lobe, lungs, and abnormality, indicating that the model relied on relevant radiological findings when making predictions.

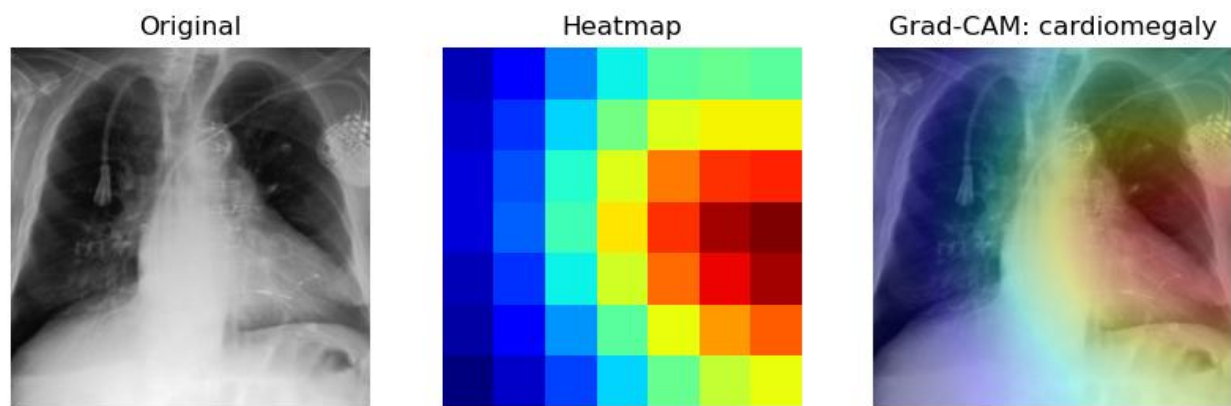


Figure 7: Original chest X-ray, activation heatmap and Grad-CAM overlay highlighting regions

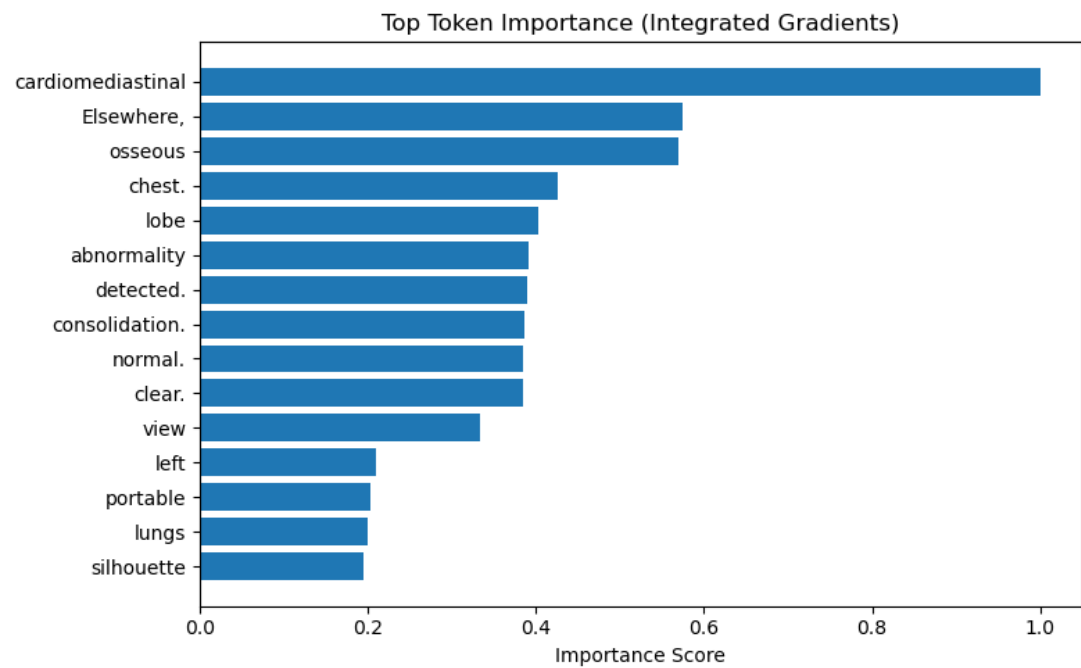


Figure 8: Token-level attribution using Integrated Gradients showing the most influential words for model predictions

Figure 8 below, presents a joint output combining the Grad-CAM heatmap with predicted disease and top attributed tokens for a single test sample. The model predicted lung opacity. Image activation is concentrated in the left lung region, indicating that the model focuses on localised areas of abnormality rather than diffuse patterns. The top-attributed tokens include opacification, pull-up, dilatation, esophagectomy, and parenchymal.

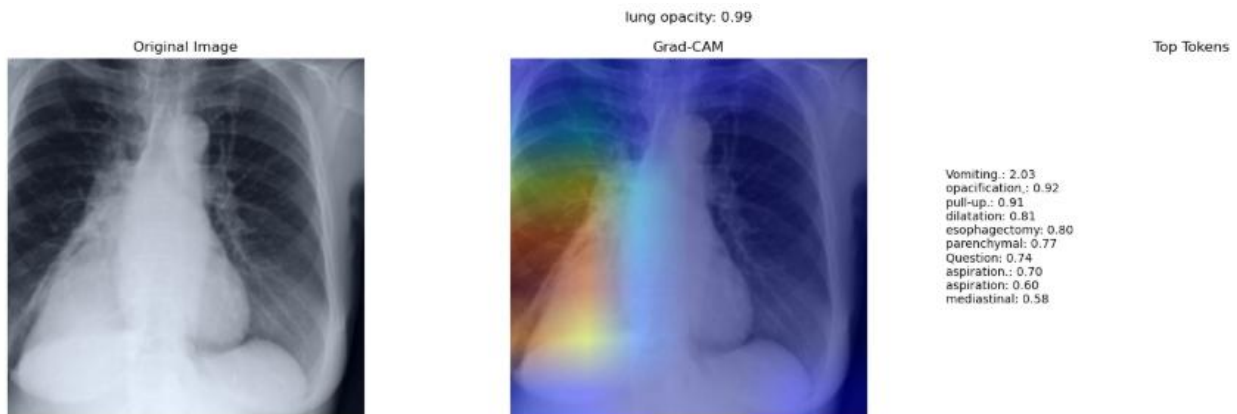


Figure 9: Joint explainability of image and text with original image, Grad-CAM, token attribution, and predicted disease.

Similar attribution patterns were observed across additional test samples (Appendix B), where Grad-CAM consistently highlighted clinically relevant thoracic regions and Integrated Gradients assigned high importance to disease-related radiological terminology.

5. DISCUSSION

This study evaluated multimodal learning, fusion strategies and explainability for chest X-ray disease classification using MIMIC-CXR. The results demonstrate that combining image and text modalities improves predictive performance while also enabling clinically interpretable explanations across both modalities.

DenseNet121 provided a competitive image-only baseline, achieving a macro AUROC of 0.8172, which is comparable to the 0.78 reported by Pillai (2022) on the same dataset. Its stronger performance on pleural effusion and edema may reflect the relatively distinct radiographic appearance of these conditions, while lower performance on pneumonia and lung opacity is consistent with previous studies reporting that these

diseases are more challenging to classify due to subtle visual manifestations, label uncertainty and class imbalance (Holste et al., 2024).

LLaMA 3.1 8B substantially outperformed DenseNet121 across all evaluation metrics, achieving a macro AUROC of 0.9177. This finding highlights the diagnostic value of radiology reports and suggests that a substantial proportion of the information required for disease classification is contained within radiologists' textual observations. Unlike image-based models that must infer disease presence from visual patterns alone, radiology reports contain expert observations and clinical reasoning produced by radiologists, providing a more direct representation of pathological findings (Danu et al., 2023). The substantial performance gap between the image-only and text-only models also helps explain why the improvements achieved through multimodal fusion were relatively modest.

Although the gains over the text-only model were small, both multimodal approaches achieved the highest overall performance, demonstrating that image and text modalities provide complementary information and add value. The consistent improvement of both multimodal models over the text-only baseline indicates that visual information still contributes useful evidence that is not fully represented within radiology reports. These findings suggest that radiology reports provided the dominant predictive signal within the dataset, while chest X-ray images contributed complementary information that further improved classification performance. This highlights the importance of multimodal learning in clinical settings, where textual and visual information may capture different aspects of disease presentation and together provide a more complete representation of patient pathology.

The performance of the proposed multimodal framework compares favourably with previous studies. Jahanian et al. (2025) reported a macro-AUC of 0.847 using RadFuse, while the best-performing model in this study achieved a macro AUROC of 0.9238. Although differences in experimental settings and label selection limit direct comparison, the improvement may be partly attributed to the use of LLaMA 3.1 8B as a modern large language model capable of generating richer contextual representations than earlier language encoders. In addition, methodological differences, including the use of disease-specific threshold optimisation and class-weighted learning, may also have influenced performance.

Late Fusion achieved highest macro AUROC (0.9238) and sensitivity and recorded the highest AUROC in five disease classes. This may reflect the advantage of preserving modality-specific strengths through independent optimisation before combining predictions at the decision level (Huang et al., 2020). In contrast, Joint Fusion achieved highest precision, specificity and F1-score, suggesting that feature-level integration enabled the model to learn richer relationships between visual and textual representations. By jointly optimising both modalities within a shared feature space, the model could exploit complementary cross-modal information before classification, potentially producing more discriminative representations and reducing false positive predictions. Statistical testing confirmed significant differences at the macro level (p

< 0.05). However, per-class DeLong analysis revealed only two statistically significant differences between the fusion strategies, with Pleural Effusion favouring Joint Fusion and Pneumonia favouring Late Fusion. This may suggest that Late Fusion is better suited to underrepresented disease classes, while Joint Fusion may benefit disease classes with stronger positive representation through feature-level interaction. The limited number of significant per-class differences suggests that the performance gap between fusion strategies was driven by relatively small improvements distributed across multiple disease classes rather than large gains within individual diseases.

The explainability results provided further insight into model behaviour. Grad-CAM heatmaps consistently highlighted clinically relevant anatomical regions rather than background structures, suggesting that image predictions were driven by meaningful radiographic features. For example, activation around the cardiac region in cardiomegaly cases aligned with the expected radiographic presentation of the disease. Similarly, Integrated Gradients assigned high importance to clinically relevant terms such as cardiomediastinal, consolidation and abnormality, indicating that the text model relied on medically meaningful language rather than spurious correlations.

The multimodal explanations further demonstrated coherent interaction between image and text modalities. In the lung opacity example, image activation was concentrated within the affected lung region, while highly attributed tokens included opacification and parenchymal, both of which are clinically associated with the predicted condition. The agreement between visual and textual evidence suggests that the model's predictions were supported by complementary information from both modalities rather than a single dominant source. Similar patterns were observed across additional examples presented in the appendix. These findings demonstrate that clinically meaningful explanations can be generated from multimodal models and support the growing view that explainability is essential for trustworthy deployment of AI systems in healthcare (Abbas et al., 2025).

LIMITATIONS AND FUTURE WORK

The multimodal analysis relied on a subset of 59,246 studies, which may affect representativeness compared to the full MIMIC-CXR dataset. In addition, only 8 of the 14 CheXpert disease labels were included, and a single frontal image was selected per study, potentially excluding diagnostic information available in additional radiographic views. Future work should extend the framework to the full dataset, disease label set and all available image views.

The framework was limited to two modalities, namely chest X-ray images and radiology reports. Incorporating additional clinical information such as electronic health records, laboratory results and patient demographics may provide a more comprehensive representation of patient status and further improve diagnostic performance.

Finally, the models were evaluated using a single dataset, which may limit generalisability to other healthcare institutions and patient populations. Future work should validate the framework on external datasets to assess robustness across different clinical settings and imaging protocols.

6. CONCLUSION

Chest X-ray interpretation remains challenging, yet many automated systems rely solely on image data and provide limited transparency in their decision-making. This study developed an explainable multimodal framework for multi-label chest X-ray disease classification using the MIMIC-CXR dataset by integrating DenseNet121 and LLaMA 3.1 8B through Late Fusion and Joint Fusion architectures, with Grad-CAM and Integrated Gradients used to provide visual and textual explanations.

The results demonstrated that multimodal learning improved performance over single-modality approaches, confirming the value of combining chest X-ray images and radiology reports for disease classification. Late fusion achieved the highest macro AUROC (0.9238) and sensitivity, while Joint Fusion achieved the highest F1-score, precision and specificity, indicating that different fusion strategies offer distinct advantages. The strong performance of the text modality further highlighted the diagnostic value of radiology reports, while the consistent improvements achieved through multimodal fusion showed that image information provided complementary evidence beyond that contained within the reports.

The explainability analysis demonstrated clinically meaningful attributions across both modalities, with multimodal explanations providing coherent visual and textual evidence to support model predictions. Overall, this work demonstrates that explainable multimodal learning can improve chest X-ray disease classification while supporting transparency and trust in AI-assisted clinical decision-making. Future work should extend the framework to the full CheXpert label set and MIMIC-CXR dataset, incorporate additional clinical modalities and evaluate performance across external datasets to assess generalisability.

REFERENCE LIST

Abbas, Q., Jeong, W. and Lee, S.W., 2025, August. Explainable AI in clinical decision support systems: a meta-analysis of methods, applications, and usability challenges. In *Healthcare* (Vol. 13, No. 17, p. 2154). MDPI.

Bhayana, R., 2024. Chatbots and large language models in radiology: a practical primer for clinical and research applications. *Radiology*, 310(1), p.e232756.

Danu, M.D., Marica, G., Karn, S.K., Georgescu, B., Mansoor, A., Ghesu, F., Itu, L.M., Suci, C., Grbic, S., Farri, O. and Comaniciu, D., 2023. Generation of radiology findings in chest x-ray by leveraging collaborative knowledge. *Procedia Computer Science*, 221, pp.1102-1109.

Dettmers, T., Pagnoni, A., Holtzman, A. and Zettlemoyer, L., 2023. Qlora: Efficient finetuning of quantized llms. *Advances in neural information processing systems*, 36, pp.10088-10115.

Geftter, W.B., Post, B.A. and Hatabu, H., 2023. Commonly missed findings on chest radiographs: causes and consequences. *Chest*, 163(3), pp.650-661.

Holste, G., Zhou, Y., Wang, S., Jaiswal, A., Lin, M., Zhuge, S., Yang, Y., Kim, D., Nguyen-Mau, T.H., Tran, M.T. and Jeong, J., 2024. Towards long-tailed, multi-label disease classification from chest X-ray: Overview of the CXR-LT challenge. *Medical Image Analysis*, 97, p.103224.

Huang, G., Liu, Z., Van Der Maaten, L. and Weinberger, K.Q., 2017. Densely connected convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 4700-4708).

Huang, S.C., Pareek, A., Seyyedi, S., Banerjee, I. and Lungren, M.P., 2020. Fusion of medical imaging and electronic health records using deep learning: a systematic review and implementation guidelines. *NPJ digital medicine*, 3(1), p.136.

Imran, M., Lee, C. and Lee, Y., 2025, September. MATEX: Multi-scale Attention and Text-Guided Explainability of Medical Vision-Language Models. In *International Workshop on Interpretability of Machine Intelligence in Medical Image Computing* (pp. 137-148). Cham: Springer Nature Switzerland.

Irvin, J., Rajpurkar, P., Ko, M., Yu, Y., Ciurea-Illcus, S., Chute, C., Marklund, H., Haghgoo, B., Ball, R., Shpanskaya, K. and Seekins, J., 2019, July. Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In *Proceedings of the AAAI conference on artificial intelligence* (Vol. 33, No. 01, pp. 590-597).

Jacenków, G., O'Neil, A.Q. and Tsaftaris, S.A., 2022, March. Indication as prior knowledge for multimodal disease classification in chest radiographs with transformers. In *2022 IEEE 19th international symposium on biomedical imaging (ISBI)* (pp. 1-5). IEEE.

Jahanian, M., Karimi, A., Eraghi, N.O. and Zarafshan, F., 2025. Multimodal transformers for joint diagnosis and retrieval from chest X-rays and radiology reports. *Informatics in Medicine Unlocked*, p.101672.

Jiang, A., 2021. *Improving Deep Learning Training and Inference with Dynamic Hyperparameter Optimization* (Doctoral dissertation, Carnegie Mellon University).

Johnson, A.E., Pollard, T.J., Berkowitz, S.J., Greenbaum, N.R., Lungren, M.P., Deng, C.Y., Mark, R.G. and Horng, S., 2019. MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports. *Scientific data*, 6(1), p.317.

Khater, O.H., Shuaibu, A.S., Haq, S.U. and Siddiqui, A.J., 2025, February. Attcdcnnet: Attention-enhanced chest disease classification using x-ray images. In *2025 IEEE 22nd International Multi-Conference on Systems, Signals & Devices (SSD)* (pp. 891-896). IEEE.

Liu, X., Pan, F., Song, H., Cao, S., Li, C. and Li, T., 2025. Mdformer: Transformer-based multimodal fusion for robust chest disease diagnosis. *Electronics*, 14(10), p.1926.

Lundberg, S.M. and Lee, S.I., 2017. A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30.

Pillai, A.S., 2022. Multi-label chest X-ray classification via deep learning. *arXiv preprint arXiv:2211.14929*.

Rajpurkar, P., Irvin, J., Zhu, K., Yang, B., Mehta, H., Duan, T., Ding, D., Bagul, A., Langlotz, C., Shpanskaya, K. and Lungren, M.P., 2017. Chexnet: Radiologist-level pneumonia detection on chest x-rays with deep learning. *arXiv preprint arXiv:1711.05225*.

Reimers, N. and Gurevych, I., 2019, November. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)* (pp. 3982-3992).

Ribeiro, M.T., Singh, S. and Guestrin, C., 2016, August. "Why should i trust you?" Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 1135-1144).

Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D. and Batra, D., 2017. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision* (pp. 618-626).

Sundararajan, M., Taly, A. and Yan, Q., 2017, July. Axiomatic attribution for deep networks. In *International conference on machine learning* (pp. 3319-3328). PMLR.

Upadhyay, J., Poudel, K. and Ranganathan, J., 2024, April. Advancing medical image diagnostics through multi-modal fusion: Insights from mimic chest x-ray dataset analysis. In *2024 IEEE 3rd International Conference on Computing and Machine Intelligence (ICMI)* (pp. 1-8). IEEE.

Voinea, Ș.V., Mămuleanu, M., Teică, R.V., Florescu, L.M., Selișteanu, D. and Gheonea, I.A., 2024. GPT-driven radiology report generation with fine-tuned Llama 3. *Bioengineering*, 11(10), p.1043.

APPENDICES

APPENDIX A — STATISTICAL SIGNIFICANCE: PER-CLASS DELONG RESULTS

This appendix presents the per-class DeLong statistical comparisons between model pairs discussed in Section 3.

Table A1: Per-class DeLong comparison between LLaMA 3.1 8B and DenseNet121.

Disease	LLaMA 3.1 8B AUROC	DenseNet121 AUROC	p-value	Significant
Atelectasis	0.8928	0.8077	5.03e-71	Yes
Cardiomegaly	0.9052	0.7919	7.41e-112	Yes
Edema	0.9391	0.8838	3.50e-34	Yes
Lung opacity	0.9043	0.7584	1.27e-165	Yes
No finding	0.9322	0.8553	1.86e-110	Yes
Pleural effusion	0.9349	0.8955	2.20e-31	Yes
Pneumonia	0.8626	0.7177	2.19e-48	Yes
pneumothorax	0.9706	0.8269	1.31e-63	Yes

To assess per-class statistical significance between LLaMA 3.1 8B (text model) and DenseNet121 (image model), DeLong comparison was applied. The results are presented in Table A1. LLaMA achieved higher AUROC across all disease classes, and all differences were statistically significant ($p < 0.05$). The largest performance gaps were observed for lung opacity, cardiomegaly, and pneumothorax, indicating that the text modality provided substantially stronger predictive information than image features alone for these disease classes.

Table A2: Per-class DeLong comparison between Late Fusion and LLaMA 3.1 8B.

Disease	Late Fusion AUROC	LLaMA 3.1 8B AUROC	p-value	Significant
Atelectasis	0.9048	0.8928	2.20e-26	Yes
Cardiomegaly	0.9145	0.9052	3.58e-21	Yes
Edema	0.9501	0.9391	1.41e-25	Yes
Lung opacity	0.9066	0.9043	1.30e-02	Yes
No finding	0.9345	0.9322	3.38e-03	Yes
Pleural effusion	0.9478	0.9349	1.49e-45	Yes
Pneumonia	0.8667	0.8626	2.15e-02	Yes
pneumothorax	0.9657	0.9706	6.49e-03	Yes

The per-class DeLong comparison between Late Fusion and LLaMA 3.1 8B in Table A2 above, shows that late fusion achieved slightly higher AUROC in seven of the eight disease classes, while LLaMA performed better only for pneumothorax. Although the AUROC differences were relatively small, all class-level differences were statistically significant ($p < 0.05$), which indicates that integrating image features with radiology report text provided measurable performance improvements over the text model across most disease classes.

Table A3: Per-class DeLong comparison between Joint Fusion and LLaMA 3.1 8B.

Disease	Joint Fusion AUROC	LLaMA 3.1 8B AUROC	p-value	Significant
Atelectasis	0.9008	0.8928	1.40e-03	Yes
Cardiomegaly	0.9111	0.9052	9.38e-03	Yes
Edema	0.9510	0.9391	5.02e-07	Yes
Lung opacity	0.9049	0.9043	7.86e-01	No
No finding	0.9333	0.9322	4.47e-01	No
Pleural effusion	0.9512	0.9349	2.16e-24	Yes
Pneumonia	0.8465	0.8626	3.25e-03	Yes
pneumothorax	0.9624	0.9706	1.40e-03	Yes

Table A3 presents the per-class DeLong comparison between Joint Fusion and LLaMA 3.1 8B. Joint Fusion achieved higher AUROC in six disease classes, while LLaMA performed better for pneumonia and pneumothorax, two of the most underrepresented classes. The largest improvements were observed for pleural effusion and edema, suggesting that integrating image features with radiology report text improved performance for disease classes with stronger positive representation. In contrast, the stronger performance of LLaMA on minority classes may reflect the increased optimisation difficulty of jointly training image and text encoders under severe class imbalance. However, differences for lung opacity and no finding were not statistically significant ($p > 0.05$). Overall, the results indicate that multimodal feature integration provided measurable performance gains over the text-only model across most disease classes.

Table A4: Per-class DeLong comparison between Late and Joint Fusion.

Disease	Late Fusion AUROC	Joint Fusion AUROC	p-value	Significant
Atelectasis	0.9048	0.9008	0.0622	No
Cardiomegaly	0.9145	0.9111	0.0839	No
Edema	0.9501	0.9510	0.6615	No
Lung opacity	0.9066	0.9049	0.3219	No
No finding	0.9345	0.9333	0.3912	No
Pleural effusion	0.9478	0.9512	0.0055	Yes
Pneumonia	0.8667	0.8465	2.20e-5	Yes
pneumothorax	0.9657	0.9624	0.1925	No

The per-class DeLong comparison between Late Fusion and Joint Fusion is presented in Table A4. Although Late Fusion achieved slightly higher AUROC in most disease classes, only pleural effusion and pneumonia showed statistically significant differences ($p < 0.05$). Joint Fusion performed better for pleural effusion, while Late Fusion achieved higher AUROC for pneumonia. Given that pneumonia was one of the most underrepresented classes, these findings may suggest that Late Fusion better preserves minority disease detection, whereas Joint Fusion may benefit disease classes with stronger positive representation. The remaining classes showed no significant differences, indicating broadly comparable performance across most disease categories.

APPENDIX B — HYPERPARAMETER TUNING

This appendix summarizes the hyperparameter tuning process conducted using the one-at-a-time (OAT) strategy described in Section 2.5. Promising parameter values identified during tuning were combined into final configurations for model training and evaluation.

Table B1: Hyperparameter Tuning for DenseNet121 Model.

Parameter	Values Tested	Best Values	Best Validation Macro AUROC
Learning rate	1e-4, 3e-5, 5e-5	3e-5	0.8125
Dropout	0.0, 0.2, 0.3	0.3	0.8125
Batch size	8, 16, 32	16	0.8114
Epochs	5,10	5	0.8143

Table B2: Final Configurations for DenseNet121 Model

Configuration	Learning Rate	Dropout	Batch Size	Epochs	Patience	Best Validation Macro AUROC
Config 1	3e-5	0.3	16	20	5	0.8116
Config 2	1e-4	0.3	16	5	3	0.8162
Config 3	3e-5	0.3	16	5	3	0.8110
Config 4	5e-5	0.3	8	5	3	0.8138

Hyperparameter tuning and final configuration selection for DenseNet121 are summarized in Tables B1 and B2. One-at-a-time (OAT) tuning identified a learning rate of 3e-5, dropout of 0.3, batch size of 16, and 5 epochs as individually strong-performing parameter values. These promising values were subsequently combined into multiple final configurations and trained for validation evaluation. Among the trained configurations, Config 2 achieved the highest validation macro AUROC of 0.8162 and was therefore selected as the final model and used for test evaluation.

Table B3: Hyperparameter Tuning for LLaMA 3.1 8B Model.

Parameter	Values Tested	Best Values	Best Validation Macro AUROC
Learning rate	1e-5, 2e-5	2e-5	0.9206
Dropout	0.05, 0.1	0.05	0.9206
Batch size	1, 2	1	0.9206

Table B3 summarises the hyperparameter tuning performed for LLaMA 3.1 8B using the OAT strategy. Due to the high computational cost of LLaMA training, only a limited number of parameter values were explored, while the baseline parameter values were also included in the evaluation. Promising parameter values identified during tuning were subsequently combined into one final configuration and evaluated. However, the baseline configuration consistently achieved the highest validation macro AUROC of 0.9206 and was therefore selected for final evaluation.

APPENDIX C — ADDITIONAL MODEL EXPLAINABILITY

This appendix presents additional explainability examples generated using Grad-CAM, Integrated Gradients, and multimodal explainability analysis.

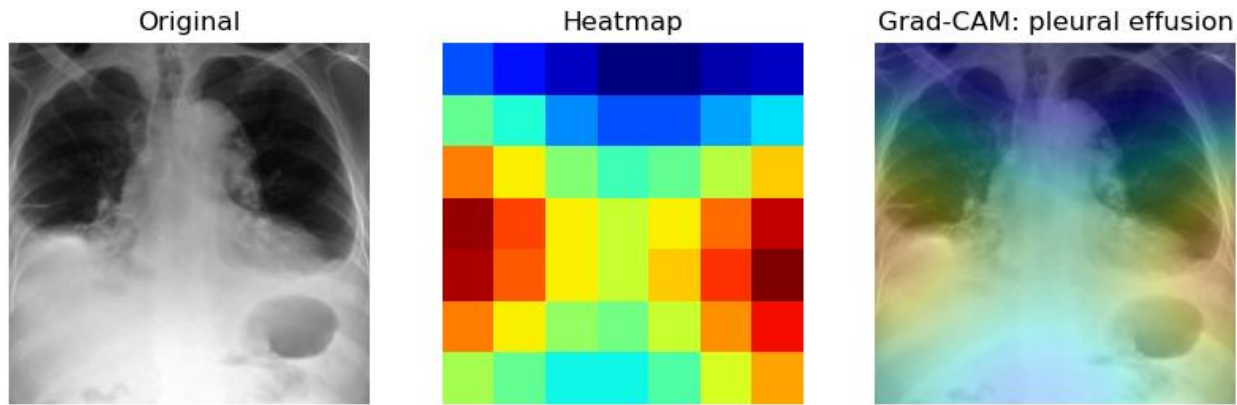


Figure C1: Grad-CAM visualisation for pleural effusion prediction. There is strong activation within the lower lung regions. This indicates that the model focused on clinically relevant anatomical areas associated with fluid accumulation.

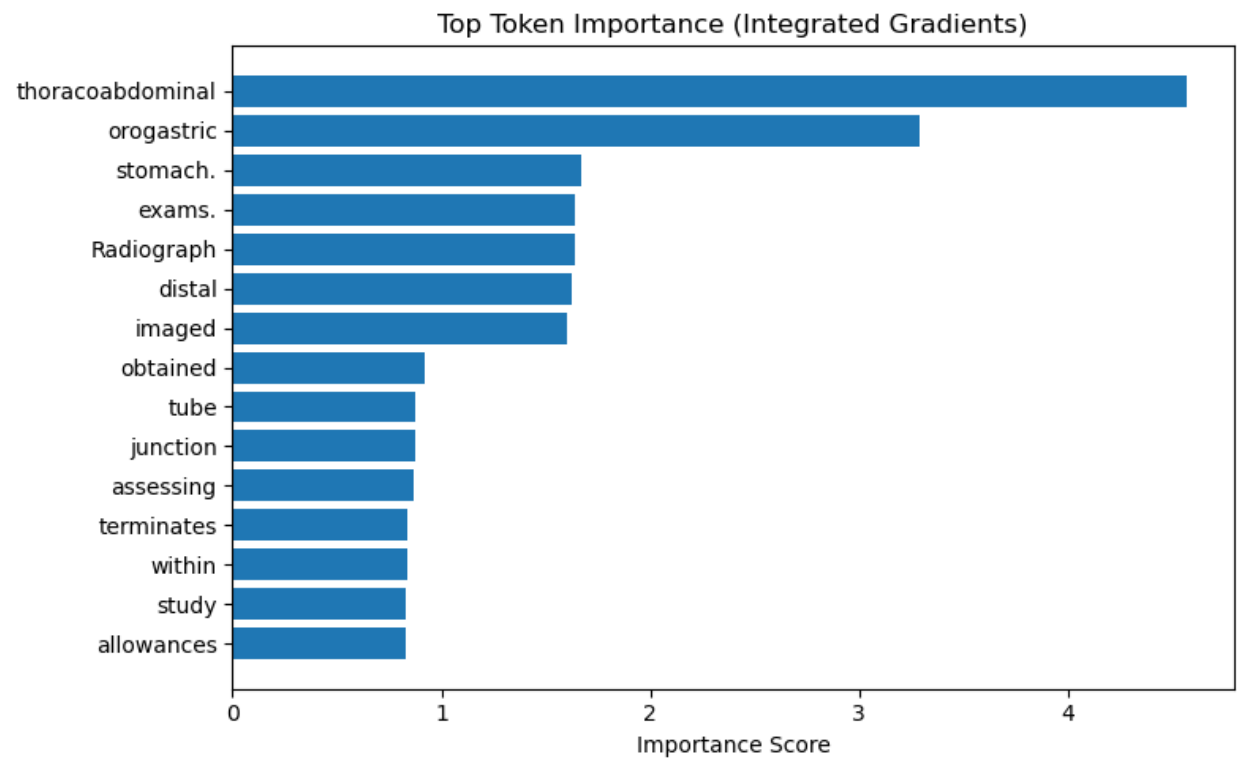


Figure C2: Integrated Gradients token attribution showing the most influential report terms that contributed to the prediction. The plot shows that tokens such as thoracoabdominal, orogastric, stomach, and tube influenced the prediction.

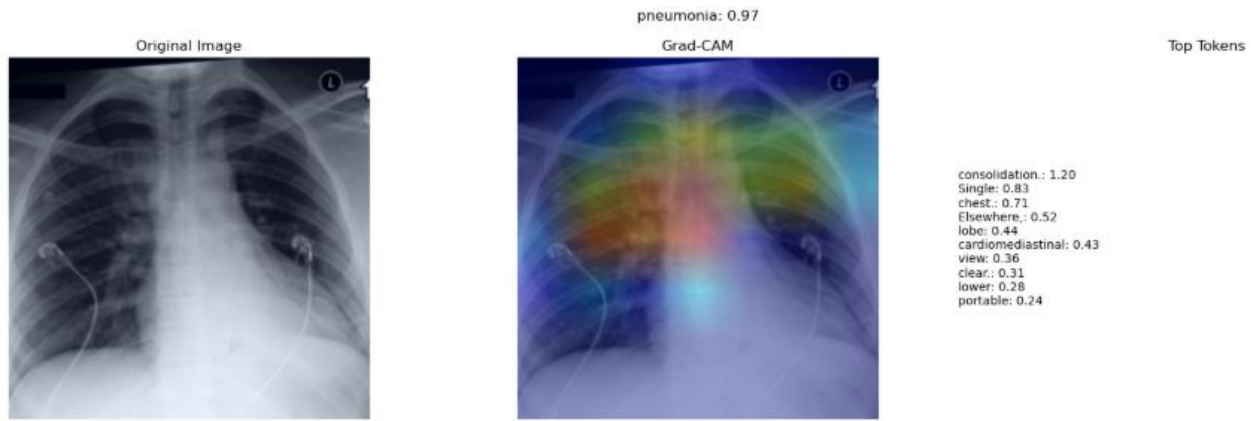


Figure C3: Joint multimodal explainability output for pneumonia prediction showing Grad-CAM activation concentrated within the lung region alongside influential report tokens such as consolidation, single, and chest. The figure demonstrates corresponding visual and textual evidence contributing to the prediction.

Additional explainability examples for image, text, and multimodal predictions are presented in Figures C1 to C3 using Grad-CAM, Integrated Gradients, and joint multimodal attribution analysis. The Grad-CAM visualization highlighted clinically relevant anatomical regions associated with pleural effusion, which indicates that the image model focused on meaningful lung areas during prediction. Integrated Gradients identified influential report terms such as thoracoabdominal, orogastric, stomach, and tube, showing that the text model relied on clinically relevant radiology language. The multimodal explanation further demonstrated consistent interaction between visual and textual evidence, where lung-region activation corresponded with influential report tokens such as consolidation. Together, these results demonstrate that the explainability framework produced clinically meaningful and interpretable predictions across image, text, and multimodal outputs.